Note Sheet of CSCI5010 – Practical Computational Geometry Algorithms, by WONG, Hok Fong

1. Random Access Machine Model (CPU + Memory)

- Set register to constant.
- Arithmetic ($+/-/\times/\div$).
- Comparison/Branching.
- Memory access (load/store).
- Randomization: **RANDINT**(x, y).

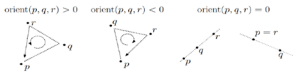
2. Real-RAM Model

- Each cell stores either an integer or real value, algorithm is forbidden to investigate a real value.
- For arithmetic operations, if one of the operands is a real value, the result is a real value.
- *floor* operation: convert x to $\lfloor x \rfloor$ given $x \in [0, 2^w)$.
- Every cell is in the range $[0, 2^w)$.
- **Constant multiplication-depth requirement:**

  Any root-to-leaf path of $T(x)$ can contain only a constant number of multiplications.

Model: $w = \Theta(\log n)$, n: input size

3. Fundamentals



$$\det \begin{pmatrix} 1 & p_x & p_y \\ 1 & q_x & q_y \\ 1 & r_x & r_y \end{pmatrix}$$

Directional vector $\pm(a, b) \Leftrightarrow$ Normal vector $(-b, a)$.

$$\text{Conv}(\{p_1, \ldots, p_k\}) = \left\{ p : \sum_{i=1}^{k} \alpha_i p_i, \text{s.t.} \sum_{i=1}^{k} \alpha_i = 1, 0 \le \alpha_i \le 1 \right\}.$$

4. Line Segment Intersection

Planesweep algorithm: maintain *partial solution on left of* ℓ, *sweepline status* (objects overlapping with ℓ), *future events on right of* ℓ.

Algorithm: Intersection Reporting (S: set of lines)

Sweepline status E: ordered dict (top to bottom)

Event queue Q: min x-coord heap (endpoints, intersections)

```
Insert all endpoints S to Q, E = ∅.
While !Q.empty() do:
  Extract event e from Q.
  s+,s−: segments immediately above/below s (endpoint), or two
         segments involved (intersection).
  Left endpoint:
    Insert segment to E.
    Remove any intersection associated with (s+,s−).
    Test (s,s+), (s,s−).
  Right endpoint:
    Delete s. Test (s+,s−).
  Intersection:
    Report e. Let (s++,s−−) be segments immediately above/below.
    Remove event associated with (s+,s++),(s−,s−−).
    Swap s+,s− in Q. Test (s−,s++),(s+,s−−).
```

Time complexity: $\mathbf{O}((n + k) \log n)$.

5. Convex Hulls

Algorithm: Graham Scan (Upper Hull)

```
Sort the points according to increasing order of their
x-coordinates, denoted (p1,p2,...,pn).
Push p1 and then p2 onto S.
For i ← 3 to n do:
    While |S| ≥ 2 and orient(pi, S[t], S[t − 1]) ≤ 0 do:
        pop S.
    Push pi onto S.
```

Invariant: Let $U(i)$ be the upper hull of $\{p_1, \ldots, p_i\}$, in step i, the stack contains (from bottom to top order) $U(i)$.

Time Complexity: $\mathbf{O}(n \log n)$

Algorithm: x-disjoint hull upper tangent $\mathbf{O}(n)$

```
p' ←rightmost point of H1, q' ← p''s predecessor.
p'' ←leftmost point of H2, q'' ← p'''s successor.
Repeat until orient(p',p'',q'') < 0 and orient(p'',p',q') > 0:
    While orient(p',p'',q') ≥ 0 do:
        advance p'',q'' to successors on H''.
    While orient(p'',p',q') ≤ 0 do:
        advance p',q' to successors on H'.
```

Chan's Algorithm: $O(n + n \log \hat{k} + \hat{k} \cdot n/\hat{k} \cdot \log \hat{k})$ (Divide, Sort Hull, Wrap Hull)

Guessing Alg.: $h^* \leftarrow 2^{2^0}, 2^{2^1}, \ldots, 2^{2^i}$. $n \log \sum h^*$, dominated by last term $k = 2^i$.

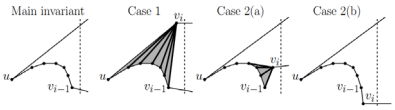
$k \leftarrow \lceil n/h^* \rceil$. Partition P into $P_1, \ldots, P_k$, and $|P_i| \le h^*$ for all i.

Compute hulls for $H(j) = \text{conv}(P_j)$ using Graham's Scan.

Find bottommost point. Apply Ex 4-2, CCW Jarvis Scan.

6. Polygon Triangulation

Monotone Polygon Triangulation: $\mathbf{O}(n)$

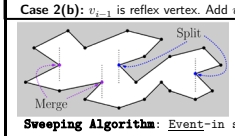


Invariant: For $i \ge 2$, let v_i be the vertex just processed by the triangulation algorithm. The untriangulated region on left of v_i consists of two x-monotone chains, a lower chain and an upper chain each containing at least one edge. If the chain from v_i to u has two or more edges, then these edges form a reflex chain. The other chain consists of a single edge whose left endpoint is u and whose right endpoint lies to the right of v_i.

Case 1: v_i lies on the opposite chain from v_{i-1}. Add diagonals joining v_i to reflex chain from v_{i-1} to u (excluding u). $u \leftarrow v_{i-1}$. Reflex chain contains only $\overline{v_i v_{i-1}}$.

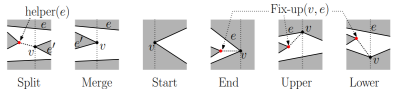
Case 2(a): Non-reflex vertex v_{i-1}, pop visible vertices till v_j, add diagonals joining v_i to these vertices (which are popped). Add v_j, v_i to the reflex chain.

Case 2(b): v_{i-1} is reflex vertex. Add v_i to the reflex chain.



helper(e_a): $e_b \leftarrow$the edge of the polygon lying just below e_a on the sweepline. The helper is the rightmost vertically visible vertex on or below e_a between e_a, e_b.

Sweeping Algorithm: Event-in sorted list (x-coord); Sweepline status-list of edges from top to bottom $\cap$ ℓ.



7. Dominance Screening

P, Q: sets of points in $\mathbb{R}^d$, report all points in Q not dominated by any points in P. Set $|n| = |P| + |Q|$.

1D - $O(n)$

2D - $O(n \log n)$, **D&C**, **obs.** right answer in result, left point q is in final result iff not dominated by right P.

Resort to 1D inst., $P' \leftarrow$project P_2 onto ℓ, $Q' \leftarrow$project A_{left} to ℓ.

Generalizable: $\mathbf{O}(n \log^{d-1} n)$. 3D - **Ex. 3-3 Planesweep** $\mathbf{O}(n \log n)$

Algorithm: Discretialization. Sort z^-, x^-, y^-. α: $z = -\infty$ to $z = +\infty$.

Fenwick tree: Report $\max_{z_i \le z \le z_0} y_i$ for Q, insert for P. **Ex. 3-4 Maxima**

8. Rectangle Point Containment

R: set of axis-parallel rectangles; P: sets of points in $\mathbb{R}^d$, report all pairs $(r, p) \in R \times P$ s.t. r contains p.

1D - $\mathbf{O}(n \log n + k)$; **kD** - remove R_{span} solve **(k-1)D** instance by projection on any one axis, split by plane in halves on the axis picked, $\mathbf{O}(n \log^{d-1} n + k)$; 2D $\mathbf{O}(n \log n + k)$: sort y-coord in adv. (**Ex. 3-1** 2D Dom.)

9. Output Sensitive Maxima

Multi-rank selection: D&C on rank list, split arr.

Algorithm: Output Sensitive Maxima

Ascending x-coordinate (for each group) $\mathbf{O}(n \log k)$.

-> 1^{st} element's rank for every group

Eliminate from n^{th} group to 1, save highest,

$$\mathbf{O}(t_{i+1} \cdot |P_{i+1}|) = \mathbf{O}(t_{i+1} \cdot n/\hat{k}), \mathbf{O}\left((n/\hat{k}) \sum_{i=1}^{k} t_i\right) = \mathbf{O}(n)$$

Lemma. *If an upper bound* $\hat{k}$ *of* k *is known, we can find all the maximal points in at most* $\mathbf{O}(n \log \hat{k})$ *time.*

10. Closest Pair

Packing Lemma: Impose a regular grid on $\mathbb{R}^d$ where every cell is a box with side length s on each dimension. Any ball with radius r can overlap with no more than $(1 + \lceil \frac{2r}{s} \rceil)^d$ cells.

Algorithm Analysis: Impose a grid, (i) each cell is axis-parallel square, $\ell = r/\sqrt{2}$, (ii) ℓ is a line in the grid. Hash bucket non-empty cells.

Obs. 1 Every cell contains at most $O(1)$ points.

Obs. 2 Every cell has at most $O(1)$ r-neighbouring cells.

Obs. 3 There are at most $2^d \cdot n = O(n)$ non-empty cells.

```
for every non-empty cell c1 on the left of ℓ do
  for every r-neighbor cell c2 of c1 on the right of ℓ do
    calculate the distance of each pair of (p1,p2) ∈ c1(P) × c2(P)
return the closest among all pairs inspected at Line 3
```

Deterministic: sort x-coord on init., investigate 40 y-neighbors, merge y-coord. (only need right part sorted)

11. Well-Separated Pair Decomposition

Define $A \otimes B = \{\{a, b\} | a \in A, b \in B, a \ne b\}$. Compute s-WSPD $\{\{A_1, B_1\}, \{A_2, B_2\}, \ldots, \{A_m, B_m\}\}$, such that (i) $A_i, B_i \in P, A_i \cap B_i = \emptyset, \forall\, i \in [m]$; (ii) $\bigcup_{i=1}^{m} A_i \otimes B_i = P \otimes P$; (iii) A_i, B_i can be enclosed within two balls of radius r s.t. closest dist between these balls is at least sr.

Algorithm: ws(u, v, s) $\mathbf{O}(n \log n + s^d n)$ time. Compressed-QT

```
if (u and v are leaves and u = v) return;
if (rep(u) or rep(v) is empty) return ∅;
else if (u and v are s-well separated) return {{u,v}};
else { if (level(u) > level(v)) swap u and v;
  Let u1,...,uk denote the children of u, return
⋃_{i=1}^{k} ws(ui, v, s);}
```

Diameter $(1 - \varepsilon)$: $s = \frac{4(1-\varepsilon)}{\varepsilon}$; t-spanner: $\frac{4(1+t)}{t-1}$, Graph: $E_i = (rep(A_i), rep(B_i))$

12. ε-kernels

Compute $Q \subseteq P$, s.t. $W(R, \vec{u}) \ge (1 - \varepsilon) W(P, \vec{u})$, $W(S, \vec{u}) = \max_{p \in S} (\vec{p} \cdot \vec{u}) - \min_{p \in S} (\vec{p} \cdot \vec{u})$. $|Q| = O((1/\varepsilon)^d)$.

Lemma. There exists an affine transformation T such that $T(P)$ is a $(1/d)$-canonical ($\alpha = \frac{\lambda_-}{\lambda_+}, \lambda_+ = \frac{1}{2}$). **John's Theorem,** $\mathbf{O}(n)$ time.

Algorithm: Quick and Dirty Kernel Constructions

```
Apply lemma, P' ← T(P). ε' = ε/d. (Fact: W(T(P),u) ≥ 1/d.)
Impose grid with side length ε'/(2√d). (Inner dist≤ ε'/2)
For every non-empty cell, take an arbitrary representative to
form R, O(n) time expected (hash).
```

13. Duality

$(p_1, p_2, \ldots, p_d) \Leftrightarrow y = p_1 x_1 + p_2 x_2 + \cdots + p_{d-1} x_{d-1} - p_d$.

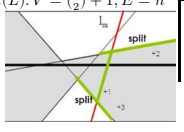
Incidence preserving/Order reversal/Intersection preserving

L to R upper hull points $\Leftrightarrow$ R to L lower envelope lines

14. Line Arrangements $\mathcal{A}(L)$: $V = \binom{n}{2} + 1, E = n^2$

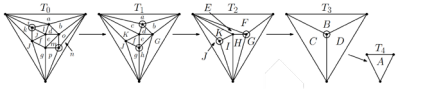
Zone Theorem: For any line ℓ, zone complexity $\le 6n$.

Induction, L-bounding edges $\le 3n$.

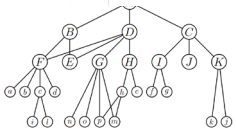


15. Point Location

Lemma. For any triangulated SLPG with n vertices, must exist S s.t. (i) S does not contain 3 dummy points; (ii) $|S| \ge (\frac{n}{2} - 3)/12$; (iii) **Every vertex in S has a degree of at most 11.**



Build: $\mathbf{O}(n \log n)$; **Query:** $\mathbf{O}(\log n)$ search each of the (at most 11) triangles in T_{k-1} that overlap this triangle. T_k to T_0. size #triangles $= \mathbf{O}(n)$. **(Ex. 11-3)**



16. Voronoi Diagram

$\bigcup_{p \in P} \boldsymbol{V}(p)$, $\boldsymbol{V}(p) = \{q \in \mathbb{R}^2 | p \text{ is a NN of } q\}$.

Thm. Voronoi diagram of a point set P has complexity $\mathbf{O}(n)$ ($n + 1$ faces, $\le 2n - 5$ vertices, $\le 3n - 6$ edges). NN-search is solved in $\mathbf{O}(\log n)$ (point location).

Fact. 1. A point q is on an edge $\Leftrightarrow \exists$ a circle centered at q passing ≥ 2 points in P and the circle is empty in the interior. 2. A point q is on the vertex $\Leftrightarrow \exists$a circle centered at q passing ≥ 3 points in P and the circle is empty in the interior. 3. Every vertex in the Voronoi diagram has a degree of 3 (assuming general position).

17. Delaunay Triangulation

Lemma. Every point set triangulation has $2n - h - 2$ edges.

Circumcircle property: circumcircle of any triangle in the Delaunay triangulation is empty.

Empty circle property: two sites p_i, p_j are connected by an edge in the Delaunay triangultion iff there is an empty circle passing through p_i and p_j. **Closest pair property**: the closest pair of sites in P are neighbors in the Delaunay triangulation.

Every edge of $\text{CH}(P)$ is an edge in $\text{DT}(P)$.

18. Backward Analysis

$\mathbf{Pr}[A] = \sum_{i=1}^{k} \mathbf{Pr}[A|B_i] \cdot \mathbf{Pr}[B_i]$. $\sum_{i=3}^{n} \mathbf{O}(1) + \mathbf{Pr}[X_i = 1] \cdot \mathbf{O}(i)$.

$\mathbf{Pr}[X_i = 1] = \sum_{\text{all } \pi} \mathbf{Pr}[X_i = 1 | \Pi = \pi] \mathbf{Pr}[\Pi = \pi] = \frac{i!}{n!} \sum_{\text{all } \pi} \mathbf{Pr}[X_i = 1 | \Pi = \pi] = \frac{i!}{n!} \sum_{\text{all } \pi} \frac{2}{i} = \frac{2}{i}$.

19. Linear Programming

Feasible: $(v_{n-1} \in h_n)$ The optimum vertex does not change.

Infeasible: $(v_{n-1} \notin h_n)$ new optimum vertex lies on the hyperplane bounding h_n.

20. Minimum Enclosing Balls

$(p_{N+1}, p_3, \ldots, p_n) \leftarrow$a random permutation of $P \backslash \{p_1\}$

$B \leftarrow$ the smallest ball covering $p_1, \ldots, p_d$

for $i = 3$ to n do

if $p_i \notin B$ then $B \leftarrow$ **M**-Points-Fixed-MEB($\{p_1, \ldots, p_i\}, \{p_1, \ldots, p_{N-1}\}$)

21. VC-dimension, ϵ-nets

Let $S = (X, \mathcal{R})$ be a range space. For $Y \subseteq X$, let $\mathcal{R}_{|Y} = \{r \cap Y, r \in \mathcal{R}\}$, if it contains all subsets of Y, then Y is shattered by $\mathcal{R}$. VC-dimension is the maximum cardinality of a shattered subset of X. If there are arbitrary large shattered subsets, then $\dim_{VC}(S) = \infty$. Halfspaces in $\mathbb{R}^d$: d+1; Axis-parallel rectangles in $\mathbb{R}^d$: 2d.

ε-net: For a $Y \subseteq X$, any $Q \in \mathcal{R}$ with $|Q \cap X| \ge \varepsilon |Q|$ implies $|Q \cap Y| \ne \emptyset$.

Sauer's Lemma: If $(X, \mathcal{R})$ is a range space with VC-dimension d and $|X| = n$, then $|R| \le \Phi_d(n)$.

22. Exercises Proof Sketch

Ex. 1-2&1-4: Batch Line Dragging Ex. 1-3: Range Max (pt-val), BBST

Ex. 1-5: Batched Point Location Ex. 1-6: Rotating Sweep

Ex. 2-1: Top-1 Search Maximality, Unimodality, Tenary Search*

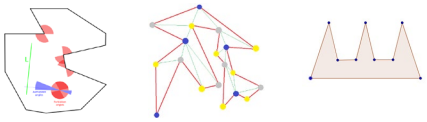
Ex. 2-2: Merging Convex Hulls

Ex. 2-3: Ear-Cutting Theorem v leftmost, 1. u, w adjacent to v, 2. v' farthest from uw.

Ex. 2-4: Triangulation Complexity

Ex. 2-5: Point in Polygon General search lowest pt, triangulate, binary sch

Ex. 2-6: Rotationally Monotone Polygon Ex. 2-7: Art Gallery Theorem



Ex. 4-1: General Binary Search

Ex. 4-2: Gift Wrap Utility

Ex. 6-3: WSPD Closest Pair $|qq'| \le 2d_{min} < sd_{min} \le |pq|$

Ex. 6-4: WSPD Graph Induction on pairs of ascending distance

Ex. 8-3: Diameter Prop. Tangent of circle **Ex. 8-4: Proof of Rotating Caliper**

Ex. 9-1: Affine Transformation

Ex. 9-2: $(1 - \varepsilon)$**-approx. Top-1 Search**

Ex. 10-1: Proof of Rotating Caliper (II)

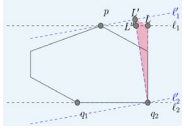
x. 10-2: Rotating Caliper Algorithm

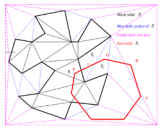
Ex. 10-3/11-1: Bounding Rect/Triangle Ex. 10-4: k**-corridor Problem**

Ex. 10-5: Smallest Area Triangle Arrangement, Sweep, Closest line to ℓ'

Ex. 11-2: Triangulating SLPG Bound triangle, Chazelle triangulation, Break hole

Ex. 11-5/6: Trapezoidal Map Complexity/Construction





Ex. 12-6: Delaunay Triangulation to Convex Hull

Ex. 13-1/2/3/4/5: LP Infeasibility+Point Crossing $\mathbf{O}(n)$ expected time

Ex. 13-6: Star-shaped Polygon

Ex. 14-1: MEB Backward Analysis "no point known" $\mathbf{O}(n)$ expected time

Ex. 14-4: 3DMEB $\mathbf{O}(n)$ expected time

Ex. 14-2: Minimum Enclosing Square Ex. 14-3: MEB Approximation

Ex. 14-5: Lowest point of circle intersection Ex. 15-1: Circumcircle Arc Covering

Ex. 15-2: Incircle Test Ex. 15-3: EMST $\mathbf{O}(n \log n)$ **Ex. 15-4: TSP Approximation**

Ex. 15-5: k**-Clustering**

Ex. 16-1: Halfplane VC-Dimension Ex. 16-2: 7-gon Supporting Line Shattering

Ex. 16-3: Sauer's Lemma Ex. 16-4/5/6/7: ε**-net size and sampling probability**